

Compiler Design (Midterm Lab 5)

Task: You are given an input file named **input.txt**. Each line contains a simple C/C++ like statement.

Step A, Token classification

Read each line and identify tokens from the following categories:

1. **Keyword**

Use only these keywords:

int, float, double, char, void, if, else, while, return

2. **Identifier**

A valid identifier:

- starts with a letter or underscore
- remaining characters may be letters, digits, or underscore

3. **Integer Constant**

A token containing only digits

4. **Operator**

Use these operators only:

+ - * / = < >

5. **Delimiter**

Use these delimiters only:

(){}[] , ;

Ignore spaces while reading.

Step B, Write classified tokens to a file

Create a file named **tokens.txt**.

For each token, write its class in this format:

Line <n>: <token> -> <class>

Step C, Build a simple symbol table

Create another file named **symbol_table.txt** containing only the **unique identifiers** found in the input.

Write them in this format:

<serial> <identifier> <first line number>

If the same identifier appears again later, do not insert it again.

Step D, Add summary

At the end of **tokens.txt**, add the total count of:

- Keywords
- Identifiers
- Integer Constants
- Operators
- Delimiters

Sample input.txt	Expected tokens.txt	Expected symbol_table.txt
<pre>int sum = a1 + 25; float x; if(x>0) return x;</pre>	<pre>Line 1: int -> Keyword Line 1: sum -> Identifier Line 1: = -> Operator Line 1: a1 -> Identifier Line 1: + -> Operator Line 1: 25 -> Integer Constant Line 1: ; -> Delimiter Line 2: float -> Keyword Line 2: x -> Identifier Line 2: ; -> Delimiter Line 3: if -> Keyword Line 3: (-> Delimiter Line 3: x -> Identifier Line 3: > -> Operator Line 3: 0 -> Integer Constant Line 3:) -> Delimiter Line 3: return -> Keyword Line 3: x -> Identifier Line 3: ; -> Delimiter Summary: Keywords = 4 Identifiers = 5 Integer Constants = 2 Operators = 3 Delimiters = 5</pre>	<pre>1 sum 1 2 a1 1 3 x 2</pre>